Dashboard Project Report

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Title: Mumbai region-wise Air Quality Indexes (2017-23) – (Tableau Public)

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1. Statement of Problem

The goal was to compare the level of pollutant molecules in the air that has affected the regions. It was to identify the areas where Monitoring stations detect higher than usual AQI in a particular year.

2. Data sourcing & preparation, source, cleaning-

Simulated a data of different regions like South, Central & western Mumbai and the years having information of PM2.5 & PM10.

Data Cleaning -

- Calculated Average PM 2.5 by creating a Calculated field and used it as a filter on years, wards and monitoring stations. It helped to overlay the comparison using KPI cards.
- Created Maximum $N0_2$ using MAX() formula & calculated percentage poor AQI from the year 2017-2023 in calculated field. This method was an important step in order further fragment the dataset into easy and visual friendly dashboard.
- Used a calculated field formula using IF ELSE condition- creating parameters like Poor, Moderate & Satisfactory to identify and separate the regions, years according to their AQI indexes.

3. Methodology

1. Used Pie-chart to display AQI distribution among regions which were declared as poor vs moderate categories, in order to fill In the two different categories, used parameters to act as legend & filters with checkbox to be applied in dashboard. Another Pie chart was made to display different types of average pollutant contribution among each filtered (check-boxed) region.
2. Created Stacked Bar-chart to show year-wise Avg. PM2.5 & PM10 category – poor or moderate, separated the colors of stacked chart to distinguish between categories from 2017-23.
3. Created a Line graph of Cleaned-calculated AVG PM 2.5 of three different regions, displayed different lines indicated rise or fall of pollutants over the years. Linked filters, Parameters & legends to further work as a joined slicer by applying it to all worksheets.

4. Created Card as a legend to showcase the percentage of AQI risen in three comparative column chart of regions – Sion, Bandra & Colaba.

4. Key insights & Recommendations

- It can be identified that region/ward- Sion continues to detect highest AQI as compared to other two wards. Line chart affirm that AVG PM2.5 has been decreased significantly, like South Mumbai (Colaba), that has gone from an average of 60 PM2.5 in 2017 to 48 PM2.5 in 2023.

- The Stacked bar chart helped to conclude that no region or ward in Mumbai as been categorized as 'Satisfactory' from 2017 to 2023. While Mumbai Central has detected the worse AQI indexes since past years. This report can be used as further research or study of importance of National parks, mangrove forests & deforestation taking place within 40-50kms of Mumbai.
- These patterns also indicate the firms, startups & unions to involve more into sustainability across the city, that benefits the health, household and livelihood of their workers.

Visualization